

Alec Martin, PhD

Curriculum Vitæ

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Education

2019 — 2025 **Ph.D. in Physics**, *The University of Missouri - Columbia*, Columbia, MO
Extragalactic Astronomy: investigating star-formation activities in distant galaxies observed by HST and JWST.

2019 — 2022 **M.Sc. in Physics**, *The University of Missouri - Columbia*, Columbia, MO

2015 — 2019 **B.Sc. in Astrophysics**, *Ohio Wesleyan University*, Delaware, OH
Astrophysics. Senior thesis: Starspot Modeling on Rapid, Differentially Rotating Stars
Minors: Mathematics

Professional Experience

2019 — 2025 **Lecturer of Astronomy**, *Indiana University - Bloomington*, Bloomington, IN
Teach introductory astronomy courses; develop research opportunities for first- and second-year undergraduates; support recruitment and retention of astronomy majors.

2020 — 2026 **Visiting Lecturer in Physics**, *Troy University*, Troy, AL
Instructor of Introductory Physics, Astronomy and their corresponding laboratories, as well as an upper-level Astrophysics course

2020 — 2025 **Research Assistant**, *University of Missouri - Columbia*, Columbia, MO

2019 — 2020 **Teaching Assistant**, *University of Missouri - Columbia*, Columbia, MO
Algebra-based introductory physics laboratory.

2017 — 2019 **Teaching Assistant**, *Ohio Wesleyan University*, Delaware, OH
Lecture introductory astronomy teaching assistant.

Research

2019 — **Ultraviolet Imaging of the Cosmic Assembly Near-infrared Deep Extragalactic Legacy Survey Fields (UVCANDELS)**
Team member of UVCANDELS - *the definitive extragalactic UV imaging of the four premier HST deep-wide survey fields best suited to JWST observations.*

- Lead study on star-formation within galaxies and properties of their star-forming substructures.
- Contribute to the efforts of expanding UVCANDELS to a larger area.

2023 — **Studying Giant Star-forming clumps with the James Webb Space Telescope**
Current work using an accepted JWST archival research proposal is designed to provide unprecedented accuracy clump stellar population property estimates (stellar mass, age, dust extinction, star formation rate) in nearby z galaxies, and extending this research to distant galaxies.

- Lead study on properties of standard source detected clumps through Bayesian SED fitting.
- Collaborate with machine learning and simulated clump research groups.

Proposals

2025 • **Securing Spectroscopic Redshifts in a Galaxy-Galaxy Lensing System in CEERS, JWST Cycle 5 GO Proposal, CO-I, Pending**

- Contributed to the writing of the Scientific Justification and Technical Justification.
- Constructed SED fitting plots.

2025 • **Securing Spectroscopic Redshift for a Candidate High-redshift Spiral Galaxy, Gemini Fast Turnaround, CO-I, Accepted**

- Contributed to the writing of the Scientific Justification.
- Awarded 3.9 hours of observing time (ID: GN-2025A-FT-206).

2023 • **Analyzing Giant Clumps in JWST Images of Star-Forming Galaxies to Constrain Feedback, JWST Cycle 2 AR Proposal, CO-I, Accepted**

- Co-I that will lead the study investigating the giant star-forming region's physical properties through observational data.
- Direct collaboration with Co-I machines learning and simulation groups.

Honors and Awards

2025 • **Ronald J. Boain PhD Dissertation Award, University of Missouri - Columbia, MO**

2025 • **245th American Astronomical Society Chambliss Award, Winner, National Harbor, MD**

2022 • **240th American Astronomical Society Chambliss Award, Honorable Mention, Pasadena, CA**

2019 • **Sigma Pi Sigma Honor Society, Inductee, Delaware, OH**

Conferences, Meetings & Talks

2025 **University of Missouri - Dissertation Defense: *Star-Forming Clumps and Their Host Galaxies: Insights from HST and JWST Observations*, Columbia, MO, April 29.**

University of Missouri - Astronomy Seminar (2020 - 2025): Bi-annual semester talks given on star-formation, galaxy-galaxy lensing, SED fitting, galaxy simulation and other extragalactic research topics at home institute.

245th American Astronomical Society Conference Poster presentation: *Unveiling Galactic Building Blocks: The Role of UV-Bright Star-Forming Clumps in Galaxy Formation*, National Harbor, MD, January 13.

2024 **Mid-American Regional Astrophysics Conference (MARAC)** Talk: *Detecting and Analyzing UV and IR Bright Star-forming Clumps at $0.5 < z < 5$* , University of Kansas, Lawrence, KS, December 8.

STScI Spring Symposium: Recipes to Regulate Star Formation at All Scales Talk: *Clump Physical Properties: Constraining power of JWST-IR filters for Low & High z Galaxies*, Baltimore, MD, April 16.

243st American Astronomical Society Conference Poster presentation: *Properties of Giant Star-forming Clumps and Their Host Galaxies Observed by HST and JWST in UVCANDELS*, New Orleans, LA, January 10.

2023 **241st American Astronomical Society Conference** Invited special session talk: *Demographics of Giant UV Star-forming Clumps in Galaxies at $0.5 \leq z \leq 1$ in UVCANDELS*, Seattle, WA, January 10.

Mid-American Regional Astrophysics Conference (MARAC) Talk: *Properties of Giant Star-forming Clumps and Their Host Galaxies Observed by HST and JWST in UVCANDELS*, Benedictine College, Atchison, KS, November 3.

2022 DAWN Winter School in Astrophysics Virtual training with programs involving data reduction and spectral energy distribution (SED) fitting. Copenhagen, DEN, February 7-11.

240th American Astronomical Society Conference Special session talk: *Demographics of Giant UV Star-forming Clumps in Galaxies at $0.5 \leq z \leq 1$ in UVCANDELS*, Pasadena, CA, June 14.

Mid-American Regional Astrophysics Conference (MARAC) Talk: *Demographics of Giant UV Star-forming Clumps in Galaxies at $0.5 \leq z \leq 1$ in UVCANDELS*, University of Arkansas, Fayetteville, AR, October 8.

2018 231st American Astronomical Society Conference. Poster presentation: *Starspot Modeling On Rapid, Differentially Rotating Stars*, Washington, DC, January 11.

2017 Summer Science Research Program Symposium Poster presentation: *Starspot Modeling On Rapid, Differentially Rotating Stars*, Ohio Wesleyan University, Delaware, OH, September 18.

Outreach

2026 Kirkwood Observatory Faculty Observer, Indiana University - Bloomington, Bloomington, IN

- Faculty supervisor to the campus, graduate student - lead observatory. The Observatory hosts weekly public viewings.

2019 2025 Laws Observatory Staff Observer, University of Missouri - Columbia, Columbia, MO

- Co-Lead graduate student for public viewing events utilizing a 16" Cassegrain Celestron telescope. Responsibilities include operating telescope and establishing other special events outside of weekly outreach including private tours for girl/boy scouts, college classes and K-12 student groups.

2024 Solar Eclipse Viewing Demonstration, Alton, MO

- Provided public viewing of our solar telescope viewing of the April 8th solar eclipse.

2023 Columbia Young Scientists (CYS) Expo, University of Missouri - Columbia, Columbia, MO

- Science Exposition for local K-8 science students.

2022 2020 Rock Bridge High School Physics Demonstration, Rock Bridge High School, Columbia, MO

- Annual physics demonstration for K-8 children in the local community.

2018 2017 Rutherford B. Hayes High School Physics Demonstration, Rutherford B. Hayes High School, Delaware, OH

- Annual physics demonstration for K-8 children in the local community.

2017 Partial Solar Eclipse Viewing Demonstration, Ohio Wesleyan University, Delaware, OH

- University tabling event showcasing pinhole box viewings of the partial solar eclipse.

Leadership & Mentoring

2020 Astronomy Research Mentor, Indiana University - Bloomington, Bloomington, IN

- Research mentor to 1st and 2nd year undergraduate students.

2020 2025 Astronomy Department Journal Club Co-Creator, University of Missouri - Columbia, Columbia, MO

- Run weekly journal club meetings for graduate students to discuss newly published academic papers in astronomy.

- 2020
2021
- Mizzou Physics and Astronomy Graduate Student Association (PAGSA)**, *Vice President*, University of Missouri - Columbia, Columbia, MO
- Internal affairs and monthly meetings helping graduate students practice presentation skills.
- 2024
2020
- Student Mentor**, *University of Missouri - Columbia*, Columbia, MO
- Served as a PHYS-ASSIST mentor for younger graduate students in physics program.
 - Served as mentor for undergraduate students in program.
- 2019
2017
- Student Mentor**, *Ohio Wesleyan University*, Delaware, OH
- Served as a mentor for younger undergraduate students in physics program.

Publications

First Author

- [1] **Martin, A.**, Guo, Y., Wang, X., et al. 2023, UV-bright Star-forming Clumps and Their Host Galaxies in UVCANDELS at $0.5 \leq z \leq 1$, ApJ, 955, 106, doi: 10.3847/1538-4357/aced3e

Co-authorship

- [7] Rutkowski, M., Zabelle, B., Hagen, T., et al. (including **Martin, A.**), 2025, Recent Star Formation in $0.5 < z < 1.5$ Quiescent Galaxies, ApJL, Volume 983 L32 doi: 10.3847/2041-8213/adbe7c
- [6] Wang, X., Teplitz H., Smith, B., et al. (including **Martin, A.**) 2025, The Lyman Continuum Escape Fraction of Star-forming Galaxies at $2.4 \leq z \leq 3.0$ from UVCANDELS, ApJ, 980, 74, doi: 10.3847/1538-4357/ada4ab
- [5] Mehta, V., Rafelski, M., Sunnquist, B., et al. (including **Martin, A.**) 2024, UVCANDELS: Catalogs of photometric redshifts and galaxy physical properties, ApJS, 275 17 doi: 10.3847/1538-4365/ad7d8f
- [4] Kuhn, V., Guo, Y., **Martin, A.**, et al. 2024, JWST Reveals a Surprisingly High Fraction of Galaxies Being Spiral-like at $0.5 \leq z \leq 4$, ApJL, 968, L15, doi: 10.3847/2041-8213/ad43eb
- [3] Sun, L., Wang, X., Teplitz, H., et al. (including **Martin, A.**) 2023, The UV luminosity function at $0.6 < z < 1$ from UVCANDELS, ApJ, 972, 8, doi: 10.3847/1538-4357/ad5540
- [2] Sattari, Z., Mobasher, B., Chartab, N., et al. (including **Martin, A.**) 2023, Fraction of Clumpy Star-Forming Galaxies at $0.5 \leq z \leq 3$ in UVCANDELS: Dependence on Stellar Mass and Environment, ApJ, 951, 147, doi: 10.3847/1538-4357/acd5d61285
- [1] Pharo, J., Guo, Y., Barro, G., et al. (including **Martin, A.**) 2022, The Dwarf Galaxy Population at $z \sim 0.7$: A Catalog of Emission Lines and Redshifts from Deep Keck Observations, ApJS 261 12, doi: 10.3847/1538-4365/ac6cdf

Proficient Skills

Programming & Scripting languages

Python (SciPy, Pandas, Numpy, Photutils, Astropy), SQL, C++, UNIX Shell scripting, Mathematica, L^AT_EX

Astronomical Data Analysis

BAGPIPES, CIGALE, EAZY, TOPCAT, Ds9, AstroImageJ, Stellarium